

Лабораторная работа №10

Основы работы с модулями ядра операционной системы

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Цель работы

Получить навыки работы с утилитами управления модулями ядра операционной системы Linux.

Ход выполнения работы

Просмотр подключённого оборудования

```
root@rishardkogengar:/nome/rkogengar#  
root@rishardkogengar:/home/rkogengar# lspci -k  
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)  
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]  
00:01.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)  
        Kernel driver in use: ata_piix  
        Kernel modules: ata_piix, ata_generic  
00:02.0 VGA compatible controller: VMware SVGA II Adapter  
        Subsystem: VMware SVGA II Adapter  
        Kernel driver in use: vmwgfx  
        Kernel modules: vmwgfx  
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controller (rev 02)  
        Subsystem: Intel Corporation PRO/1000 MT Desktop Adapter  
        Kernel driver in use: e1000  
        Kernel modules: e1000  
00:04.0 System peripheral: InnoTek Systemberatung GmbH VirtualBox Guest Service  
        Kernel driver in use: vboxguest  
        Kernel modules: vboxguest  
00:05.0 Multimedia audio controller: Intel Corporation 82801AA AC'97 Audio Controller (rev 01)  
        Subsystem: Dell Device 0177  
        Kernel driver in use: snd_intel8x0  
        Kernel modules: snd_intel8x0  
00:06.0 USB controller: Apple Inc. KeyLargo/Intrepid USB  
        Kernel driver in use: ohci-pci  
00:07.0 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)  
        Kernel driver in use: piix4_smbus  
        Kernel modules: i2c_piix4  
00:0b.0 USB controller: Intel Corporation 82801FB/FBM/FR/FW/FRW (ICH6 Family) USB2 EHCI Controller  
        Kernel driver in use: ehci-pci  
00:0d.0 SATA controller: Intel Corporation 82801HM/HEM (ICH8M/ICH8M-E) SATA Controller [AHCI mode]  
        (rev 02)  
        Kernel driver in use: ahci  
        Kernel modules: ahci  
root@rishardkogengar:/home/rkogengar#
```

Список загруженных модулей

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# lsmod | sort
```

ac97_bus	12288	1	snd_ac97_codec
ahci	57344	3	
ata_generic	16384	0	
ata_piix	45056	1	
cdrom	90112	2	isofs,sr_mod
crc32c_intel	16384	0	
crc32_pclmul	12288	0	
crct10dif_pclmul	12288	1	
dm_log	24576	2	dm_region_hash,dm_mirror
dm_mirror	28672	0	
dm_mod	245760	9	dm_multipath,dm_log,dm_mirror
dm_multipath	53248	0	
dm_region_hash	28672	1	dm_mirror
drm_ttm_helper	16384	2	vmwgfx
e1000	200704	0	
fuse	253952	5	
ghash_clmulni_intel	16384	0	
i2c_piix4	36864	0	
i2c_smbus	20480	1	i2c_piix4
intel_pmc_core	126976	0	
intel_rapl_common	53248	1	intel_rapl_msr
intel_rapl_msr	20480	0	
intel_uncore_frequency_common	16384	0	
intel_vsec	20480	1	intel_pmc_core
ip_set	69632	0	
isofs	69632	1	
joydev	28672	0	
libahci	69632	1	ahci
libata	512000	4	ata_piix,libahci,ahci,ata_generic
loop	45056	0	
Module	Size	Used by	

Проверка и загрузка модуля ext4

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# lsmod | grep ext4  
root@rishardkogengar:/home/rkogengar# modprobe ext4  
root@rishardkogengar:/home/rkogengar# lsmod | grep ext4  
ext4                1187840    0  
mbcache              16384     1  ext4  
jbd2                  217088     1  ext4  
root@rishardkogengar:/home/rkogengar# modinfo ext4  
filename:            /lib/modules/6.12.0-55.12.1.el10_0.x86_64/kernel/fs/ext4/ext4.ko.xz  
softdep:             pre: crc32c  
license:             GPL  
description:         Fourth Extended Filesystem  
author:              Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and other  
s  
alias:               fs-ext4  
alias:               ext3  
alias:               fs-ext3  
alias:               ext2  
alias:               fs-ext2  
rhelversion:         10.0  
srcversion:          CBA9BD0FC931061AEC0A8F4  
depends:              jbd2,mbcache  
intree:              Y  
name:                ext4  
retpoline:           Y  
vermagic:            6.12.0-55.12.1.el10_0.x86_64 SMP preempt mod_unload modversions  
sig_id:              PKCS#7  
signer:              Rocky kernel signing key  
sig_key:             57:BC:A1:34:94:0D:57:12:99:67:28:D8:C0:8B:1C:4B:8C:8A:13:06  
sig_hashalgo:        sha256  
signature:           05:D7:3D:A0:55:FA:4A:16:70:78:23:1F:C7:9A:17:C6:A9:9C:85:71:
```

Рис. 3: Загрузка и проверка модуля ext4

Выгрузка модулей ext4 и xfs

```
srcversion:      CBA9BD0FC931061AEC0A8F4
depends:          jbd2,mbcache
intree:          Y
name:            ext4
retpoline:       Y
vermagic:        6.12.0-55.12.1.el10_0.x86_64 SMP preempt mod_unload modversions
sig_id:          PKCS#7
signer:          Rocky kernel signing key
sig_key:         57:BC:A1:34:94:0D:57:12:99:67:28:D8:C0:8B:1C:4B:8C:8A:13:06
sig_hashalgo:    sha256
signature:       05:D7:3D:A0:55:EA:4A:16:70:78:23:1F:C7:9A:17:C6:A9:9C:B5:71:
                  6B:EB:97:2C:94:5A:49:CB:07:A0:85:2E:27:8D:C4:01:27:43:2A:C8:
                  B8:37:9C:AB:EF:B7:D9:AC:70:98:F6:2B:1B:A6:9D:7D:62:0E:8C:2E:
                  4E:D6:0E:3F:B9:33:E7:D9:8E:05:F0:70:09:81:08:5F:1E:26:99:E4:
                  C2:33:CD:4A:01:B2:F3:17:11:06:81:B2:F8:C6:23:85:76:76:43:9B:
                  00:78:01:A7:51:FA:AC:9F:25:39:88:1C:FF:27:C2:AF:0B:30:64:BA:
                  D6:FB:2C:81:43:AA:61:09:3E:6B:C8:5C:43:D2:B1:BB:8F:E4:99:95:
                  45:FE:EF:EC:B8:E0:3F:A1:50:E8:31:E2:8C:F2:5D:EF:15:87:2D:54:
                  98:24:93:D0:B7:A8:17:6B:E7:5A:23:6B:DB:9D:1C:CC:42:F1:5B:2B:
                  D5:48:EE:F5:A4:EA:FC:0E:D4:AD:8B:FB:A9:0B:53:DD:33:AD:DA:F4:
                  73:F5:28:0B:2B:19:84:75:BB:86:D5:2E:34:3E:89:53:FB:FD:78:C6:
                  FB:83:F0:1D:75:8B:8C:96:8D:87:62:27:B0:58:99:A0:A0:23:E2:A7:
                  2B:E7:E7:25:2C:AB:D2:68:A2:C9:44:30:39:FF:46:82:CC:80:DF:CB:
                  3A:10:6D:7F:43:6F:35:78:A0:96:EA:BC:3C:31:E4:5F:15:DB:8E:C4:
                  01:8F:DB:E7:F9:0D:AF:80:8F:C2:59:13:7D:01:5E:78:18:AE:ED:F8:
                  FA:7A:C2:3E:9E:63:B4:57:50:92:28:E8:95:2D:D3:44:C5:8B:F5:C8:
                  0A:BE:F5:96:B5:D8:DE:8F:91:DA:85:E5:BF:A2:7D:9B:2A:4C:AD:4E:
                  07:54:52:A9:F2:62:B6:D7:58:9F:2E:8D:6A:C2:F4:8E:98:32:EB:14:
                  E6:BE:EE:3B:C5:98:36:95:53:D8:61:BD:41:8A:FF:4B:AC:49:70:B2:
                  19:94:AF:2A
```

```
root@rishardkogengar:/home/rkogengar#
```

```
root@rishardkogengar:/home/rkogengar#
```

```
root@rishardkogengar:/home/rkogengar# modprobe -r ext4
```

```
root@rishardkogengar:/home/rkogengar# modprobe -r xfs
```

```
modprobe: FATAL: Module xfs is in use.
```

```
root@rishardkogengar:/home/rkogengar#
```

Загрузка модулей ядра с
параметрами

Проверка и загрузка bluetooth

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# lsmod | grep bluetooth  
root@rishardkogengar:/home/rkogengar# modprobe bluetooth  
root@rishardkogengar:/home/rkogengar# lsmod | grep bluetooth  
bluetooth          1114112  0  
rfkill              40960  4 bluetooth  
root@rishardkogengar:/home/rkogengar# modinfo bluetooth  
filename:           /lib/modules/6.12.0-55.12.1.el10_0.x86_64/kernel/net/bluetooth/bluetooth.ko.xz  
alias:              net-pf-31  
license:            GPL  
version:            2.22  
description:        Bluetooth Core ver 2.22  
author:             Marcel Holtmann <marcel@holtmann.org>  
rhelversion:        10.0  
srcversion:         5F9AA895ADC3BA3840211D1  
depends:             rfkill  
intree:             Y  
name:               bluetooth  
retpoline:          Y  
vermagic:           6.12.0-55.12.1.el10_0.x86_64 SMP preempt mod_unload modversions  
sig_id:             PKCS#7  
signer:             Rocky kernel signing key  
sig_key:            57:BC:A1:34:94:0D:57:12:99:67:28:D8:C0:8B:1C:4B:8C:8A:13:06  
sig_hashalgo:       sha256  
signature:          2E:24:0B:EE:9E:E6:A3:58:91:2A:D3:87:25:A3:1B:46:06:89:06:BC:  
                    4A:99:10:8D:7A:0F:1E:FF:92:59:FD:59:74:DB:7E:42:40:14:03:C9:  
                    43:D0:A3:4F:F3:10:71:07:4D:FF:34:52:96:1A:8B:90:9C:DF:D8:FB:  
                    07:E9:07:16:2E:8D:4B:23:0C:92:93:0B:88:E9:82:6B:06:1B:BD:25:  
                    69:7D:37:70:CC:C2:A5:08:85:C9:34:D9:69:1C:28:BB:81:B0:8A:C7:  
                    9F:AD:5D:46:AB:C1:B2:B1:C4:E3:8B:1A:12:5D:E2:52:D0:42:DD:FD:  
                    2B:D0:5A:51:5C:6E:AD:0D:FD:90:EB:62:89:91:D2:C8:4A:25:EA:9D:  
                    CA:B9:BE:27:C7:BB:85:D6:A5:A1:7B:8B:DE:E2:CC:26:F3:D6:D1:D2:
```

Выгрузка модуля bluetooth

```
sig_id:          PKCS#7
signer:          Rocky kernel signing key
sig_key:         57:BC:A1:34:94:0D:57:12:99:67:28:D8:C0:8B:1C:4B:8C:8A:13:06
sig_hashalgo:    sha256
signature:       2E:24:0B:EE:9E:E6:A3:58:91:2A:D3:87:25:A3:1B:46:06:89:06:BC:
                  4A:99:10:8D:7A:0F:1E:FF:92:59:FD:59:74:DB:7E:42:40:14:03:C9:
                  43:D0:A3:4F:F3:10:71:07:4D:FF:34:52:96:1A:8B:90:9C:DF:D8:FB:
                  07:E9:07:16:2E:8D:4B:23:0C:92:93:0B:88:E9:82:6B:06:1B:BD:25:
                  69:7D:37:70:CC:C2:A5:08:85:C9:34:D9:69:1C:28:BB:81:B0:8A:C7:
                  9F:AD:5D:46:AB:C1:B2:B1:C4:E3:8B:1A:12:5D:E2:52:D0:42:DD:FD:
                  2B:D0:5A:51:5C:6E:AD:0D:FD:90:EB:62:89:91:D2:C8:4A:25:EA:9D:
                  CA:B9:BE:27:C7:BB:85:D6:A5:A1:7B:8B:DE:E2:CC:26:F3:D6:D1:D2:
                  8C:FF:82:18:76:56:0C:DE:85:BB:B2:BD:8C:56:99:B6:21:B5:3A:ED:
                  15:E8:CC:3D:5F:C9:84:65:1C:94:DD:98:ED:03:D3:6D:DE:F3:B7:31:
                  19:34:07:1F:19:4A:82:C6:F5:82:F3:3C:87:38:F3:A7:69:66:0E:3B:
                  54:25:78:36:FC:A5:17:C0:5A:62:C5:4C:41:F6:FF:87:5F:0C:58:45:
                  AF:12:D3:12:5D:11:49:20:BD:B1:E1:A4:DB:BA:D3:18:B6:EE:D0:57:
                  EC:75:3A:A6:0D:E2:36:37:78:C6:BB:DD:9B:1C:67:EB:D2:A8:9F:16:
                  2B:8A:CF:8F:23:38:B4:7A:A9:92:49:86:D7:9F:08:7D:8A:92:7D:DD:
                  41:D6:4B:31:F6:45:31:D4:7E:6E:E1:D6:94:BA:D0:6A:12:CC:76:A2:
                  12:55:EF:6B:91:CB:45:F0:F5:75:0B:DD:7D:6B:D7:E1:B0:13:51:40:
                  29:D2:1D:3B:DD:E2:BF:AC:4B:77:67:BA:C8:95:16:41:00:9A:8F:96:
                  21:DC:9A:C2:B2:20:31:5C:EC:90:EA:B0:24:C3:5D:B9:72:31:B8:98:
                  DB:A9:4D:FE
parm:            disable_esco:Disable eSCO connection creation (bool)
parm:            disable_ertm:Disable enhanced retransmission mode (bool)
parm:            enable_ecred:Enable enhanced credit flow control mode (bool)
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar# modprobe -r bluetooth
root@rishardkogengar:/home/rkogengar# █
```

Проверка версии и списка пакетов ядра

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# uname -r  
6.12.0-55.12.1.el10_0.x86_64  
root@rishardkogengar:/home/rkogengar# dnf list kernel  
Rocky Linux 10 - BaseOS                               4.0 kB/s | 4.3 kB      00:01  
Rocky Linux 10 - AppStream                             17 kB/s | 4.3 kB      00:00  
Rocky Linux 10 - Extras                               146 B/s | 3.1 kB      00:21  
Installed Packages  
kernel.x86_64                                         6.12.0-55.12.1.el10_0    @anaconda  
Available Packages  
kernel.x86_64                                         6.12.0-55.37.1.el10_0    baseos  
root@rishardkogengar:/home/rkogengar#
```

Рис. 7: Проверка версии ядра и доступных пакетов

```
xorg-x11-server-Xwayland-24.1.5-4.el10_0.x86_64  
yum-4.20.0-14.el10_0.rocky.0.1.noarch  
zlib-ng-compat-2.2.3-1.el10.rocky.0.1.x86_64  
zlib-ng-compat-devel-2.2.3-1.el10.rocky.0.1.x86_64
```

Installed:

```
kernel-6.12.0-55.37.1.el10_0.x86_64  
kernel-core-6.12.0-55.37.1.el10_0.x86_64  
kernel-devel-6.12.0-55.37.1.el10_0.x86_64  
kernel-modules-6.12.0-55.37.1.el10_0.x86_64  
kernel-modules-core-6.12.0-55.37.1.el10_0.x86_64  
kernel-modules-extra-6.12.0-55.37.1.el10_0.x86_64  
libatomic-14.2.1-7.el10.x86_64  
libdex-0.8.1-1.el10.x86_64
```

Complete!

```
root@rishardkogengar:/home/rkogengar# dnf update kernel
```

Last metadata expiration check: 0:04:07 ago on Wed 15 Oct 2025 02:38:55 PM MSK.

Dependencies resolved.

Nothing to do.

Complete!

```
root@rishardkogengar:/home/rkogengar# dnf update
```

Last metadata expiration check: 0:04:11 ago on Wed 15 Oct 2025 02:38:55 PM MSK.

Dependencies resolved.

Nothing to do.

Complete!

```
root@rishardkogengar:/home/rkogengar# dnf upgrade --refresh
```

Rocky Linux 10 - BaseOS	9.8 kB/s 4.3 kB	00:00
Rocky Linux 10 - AppStream	16 kB/s 4.3 kB	00:00
Rocky Linux 10 - Extras	11 kB/s 3.1 kB	00:00

Dependencies resolved.

Nothing to do.

Complete!

```
root@rishardkogengar:/home/rkogengar#
```

Проверка новой версии ядра

```
rkogengar@rishardkogengar:~$ uname -r
6.12.0-55.37.1.el10_0.x86_64
rkogengar@rishardkogengar:~$ hostnamectl
  Static hostname: rishardkogengar.localdomain
        Icon name: computer-vm
        Chassis: vm 🖥️
        Machine ID: 9b971042516b4d0cb2d9a4048b7180f9
        Boot ID: 6f6aba5beb754348881fbfa7b2310f53
  Virtualization: oracle
  Operating System: Rocky Linux 10.0 (Red Quartz)
        CPE OS Name: cpe:/o:rocky:rocky:10::baseos
        OS Support End: Thu 2035-05-31
  OS Support Remaining: 9y 7month 2w
        Kernel: Linux 6.12.0-55.37.1.el10_0.x86_64
        Architecture: x86-64
        Hardware Vendor: innotek GmbH
        Hardware Model: VirtualBox
  Firmware Version: VirtualBox
        Firmware Date: Fri 2006-12-01
        Firmware Age: 18y 10month 2w
rkogengar@rishardkogengar:~$ █
```

Итоги работы

В ходе работы были изучены команды управления модулями ядра Linux, их загрузка, выгрузка и просмотр параметров.

Также выполнено обновление системы и ядра, что позволило закрепить навыки администрирования в среде Rocky Linux.